

Week 7, October 23, 2024

Main topics

- Iterative Methods for Large-Scale Problems
- Generalizations of Tikhonov regularization

Material

From last week: P. C. Hansen, *Discrete Inverse Problems: Insight and Algorithms*, Chapter 6 supplemented by Kirsch, section 2.6.

New Material: P. C. Hansen, *Discrete Inverse Problems: Insight and Algorithms*, Chapter 8.1 and 8.6.

Program

13:00-13:30 Iterative Methods for Large-Scale Problems

13:30-15:45 Exercises (see below)

16:00-17:00 Lecture: Generalized Tikhonov regularization

Exercises

1. The purpose of this exercise is to show how that, for a fixed α , we can compute the same Tikhonov solution by means of a factorization method (in Matlab: `backslash`, which uses a QR factorization) and an iterative method (in Matlab: `lsqr` which is a very robust implementation of CGLS). Recall that the least squares formulation of the Tikhonov problem is

$$\min_x \left\| \begin{pmatrix} A \\ \sqrt{\alpha} I \end{pmatrix} x - \begin{pmatrix} y \\ 0 \end{pmatrix} \right\|_2. \quad (1)$$

We use the `deblurring` model problem, and to simulate a large-scale problem we use $n = 2500$ (we use this approach to avoid implementing a realistic large-scale problem in two dimensions). Add Gaussian noise with standard deviation $\eta = 0.01$ to the exact right-hand side: $y^\delta = y + \text{eta} * \text{randn}(n, 1)$. We recommend that you use $\alpha = 2.5\text{e-}3$ (feel free to try other values).

Now solve the Tikhonov problem in the formulation (1) with these two methods:

- QR factorization – Matlab does this, behind your back, when you use `backslash`.
- The iterative method CGLS – in Matlab we use the build-in function `lsqr` which is a very robust implementation of CGLS.

Can you confirm that the two computed solutions are practically identical?

Determine the computing times for the two methods. In Matlab you can use `tic` and `toc`. Which is faster? (If you have patience you can also try larger values of n .)

2. We illustrate the use of the Landweber iteration with a projection,

$$x^{[k]} = P(x^{[k-1]} + \omega A^T(y - Ax^{[k-1]})) ,$$

where P represents a nonnegativity constraint:

$$\text{if } x \text{ and } y = P(x) \text{ are vectors, then } y_i = \begin{cases} x_i, & x_i \geq 0 \\ 0, & x_i < 0. \end{cases}$$

This method converges to a non-negative least squares solution.

We use the `Abel` test problem with the non-negative function

$$x(s) = s^2 \cos(\pi/2s).$$

Implement two versions of Landweber's method: the standard version and the version with projection. Use both methods to solve the test problem with $n = 64$ and **no noise** in the data; you may need several hundred iterations; experiment with the stepsize, e.g. $\omega = 1.5/\text{norm}(A)^2$. Compute the error histories for the two methods, study the regularized solutions, and comment on the results.

3. Now add noise with standard deviation $\eta = 0.3$ to the right-hand side, and run the experiments again. Do you see semi-convergence? If you do, at which iteration do we obtain the smallest error, for the two methods?
4. Run the problem above with `cgl`s instead of Landweber. **MA**ke a plot of the error history and semi-convergence.
5. Consider Figure 6.6 in *Discrete Inverse Problems: Insight and Algorithms*. Take the Abel model problem and perform a similar analysis.
6. Iterative methods are particularly relevant for large scale problems. But how large is large? Take $N = 2^n$ for $n = 5, 2, \dots$ and generate a random matrix $A \in R^{N \times N}$. For how large an N can you in reasonable time (seconds)

(a) Compute the SVD?

(b) Solve the Tikhonov problem (for random y)

$$\|Ax - y\|_2^2 + \alpha\|x\|_2^2 = \left\| \begin{bmatrix} A \\ \sqrt{\alpha}I \end{bmatrix} x - \begin{bmatrix} y \\ 0 \end{bmatrix} \right\|_2^2.$$

7. Implement the 2D forward heat problem on $(0, \pi)^2$ with homogeneous Dirichlet boundary conditions using FFT etc. The operator takes an image $x(s_1, s_2)$ and produces

$$y(t_1, t_2) = Kx(t_1, t_2) = \int k(t, s)x(s) ds$$

with the kernel

$$k(t, s) = \sum_{n, m} e^{-(n^2 + m^2)T} \sin(ns_1) \sin(ms_2) \sin(nt_1) \sin(mt_2).$$